

THE BLACK BOX — AI-Powered Telegram Bot

A cryptic, mysterious Telegram bot powered by Groq's Llama 3.3 70B. Responds to questions as if it sees patterns hidden from ordinary people.

Architecture

Layer	Technology
AI Engine	Groq API (Llama 3.3 70B)
Telegram	python-telegram-bot v20
Web Server	Flask + Gunicorn
Hosting	Render (free tier)
Fallback	Deterministic keyword engine

Prerequisites

1. **Telegram Bot Token** — Get from [@BotFather](#)
 2. **Groq API Key** — Sign up free at [console.groq.com](#)
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Deploy to Render

Option A: One-Click (with GitHub)

1. Push these files to a GitHub repo
2. Go to [render.com](#) → New → Blueprint
3. Connect your repo
4. Set environment variables in dashboard:
 - BOT_TOKEN — your Telegram token
 - WEBHOOK_URL — your Render app URL (e.g., <https://blackbox-bot-ai.onrender.com>)
 - GROQ_API_KEY — your Groq API key
 - GROQ_MODEL — llama-3.3-70b-versatile (default) or llama-3.1-8b-instant for higher volume

Option B: Manual Upload

1. Create new Web Service on Render
2. Upload these files
3. Set the same environment variables
4. Deploy

Set Webhook

After deployment, set the webhook manually:

```
https://api.telegram.org/bot<YOUR_BOT_TOKEN>/setWebhook?url=https://your-app.onrender.com/
```

Or add WEBHOOK_URL to Render env vars and redeploy.

Free Tier Limits (Groq)

Model	RPM	RPD	TPM
llama-3.3-70b-versatile	30	1,000	12,000
llama-3.1-8b-instant	30	14,400	6,000

For a low-traffic Telegram bot, the 70B model's 1,000 requests/day is more than sufficient. Switch to 8B if you need higher volume.

How It Works

The Cage (System Prompt)

The AI is constrained by a fortress-grade system prompt that explicitly forbids:

- Self-explanation or reasoning disclosure
- AI/ML terminology
- Emotional or supportive tone
- Lists, bullet points, long essays
- Generic self-help advice
- Breaking character

Safety Net

If the AI ever violates these rules (e.g., says “I’m an AI” or “I recommend”), the bot automatically falls back to the deterministic keyword engine — the original hardcoded BLACK BOX logic.

Response Parameters

- `temperature: 0.7` — Controlled unpredictability
 - `max_tokens: 150` — Forces brevity (1-4 sentences)
 - `frequency_penalty: 0.3` — Discourages repetition
 - `presence_penalty: 0.4` — Encourages novel phrasing
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Bot Commands

Command	Description
<code>/start</code>	Initialize contact
<code>/void</code>	Enter the Void mode

Local Testing

```
pip install -r requirements.txt
export BOT_TOKEN="your_token"
export GROQ_API_KEY="your_key"
export WEBHOOK_URL="http://localhost:5000"
python bot.py
```

For local testing, modify `bot.py` to use polling instead of webhooks.

Files

- `bot.py` — Core engine with AI + fallback
 - `requirements.txt` — Dependencies
 - `render.yaml` — Render deployment config
 - `README.md` — This file
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The system is active. You may transmit. It does not guarantee clarity.